



A multidisciplinary, symptom-targeted approach to rehabilitation in pediatric concussion

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ABSTRACT

Concussion, or mild traumatic brain injury, is a common functional brain injury with heterogeneous symptoms affecting physical, cognitive, emotional, visual, vestibular, and autonomic domains. Although most children and adolescents recover within several weeks, a substantial minority develop persistent symptoms that interfere with school participation, physical activity, and quality of life. This review summarizes principles of multidisciplinary assessment and rehabilitation for prolonged or complex concussion recovery, using the clinical model of the Multidisciplinary Concussion Clinic at Boston Children's Hospital as a practical framework. Persistent post-concussion symptoms often reflect multiple interacting impairments rather than a single unresolved process. Multidisciplinary assessment helps identify vestibular, visual, pain related, cervical, cognitive, and exertional contributors to prolonged recovery, improving patient experience and allowing more efficient, less fragmented care. Key management strategies include graded sub-symptom aerobic exercise, vestibular and oculomotor rehabilitation, treatment of post-traumatic headache and cervical pain, vision-based interventions, and individualized return-to-learn planning. Persistent post-concussion symptoms are best approached as a complex clinical problem requiring active, symptom-targeted, multidisciplinary care that can streamline recommendations and treatment planning while supporting functional recovery.

Introduction

Concussion, or mild traumatic brain injury, is a common functional brain injury characterized by transient neurologic dysfunction and heterogeneous symptoms that may involve physical, cognitive, emotional, visual, vestibular, and autonomic domains. In children and adolescents, concussion is especially important because symptoms can interfere with school participation, physical activity, social functioning, and quality of life. Although most patients recover within several weeks, a substantial minority develop persistent post-concussion symptoms (PPCS) that prolong recovery and complicate return to normal activities.¹ Historically, management emphasized rest and gradual return to activity. More recent clinical guidance has shifted away from prolonged

rest toward early education, symptom-limited activity, and active rehabilitation tailored to the patient's clinical profile.² Individuals with persisting post concussion symptoms, typically defined as symptom duration greater than 4 weeks, should undergo a multimodal clinical assessment that includes standardized and validated symptom rating scales.³

Persistent symptoms after concussion often do not reflect a single ongoing process. Rather, they may arise from multiple interacting contributors, including vestibular, visual, headache, cervical, cognitive, emotional, sleep-related, and exertional factors. The value of multidisciplinary assessment is reflected in prior clinical experience from a pediatric tertiary-care multidisciplinary concussion clinic, in which children evaluated for PPCS were found to have a wide range of

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specialty-specific, potentially treatable diagnoses. The most common included headaches, neck pain, sleep disturbances, binocular vision dysfunction, anxiety, depression, refractive error, myofascial pain syndrome, and benign paroxysmal positional vertigo.⁴ These findings support the view that prolonged recovery is often shaped by multiple coexisting impairments rather than a single unresolved injury and help explain why single-specialty care may be insufficient. Multidisciplinary assessment can help identify the principal contributors to ongoing symptoms, improve patient experience, and provide more efficient, less fragmented care while streamlining recommendations and treatment planning.

In the Multidisciplinary Concussion Clinic (MDCC) at Boston Children's Hospital, evaluations are completed in a single coordinated day, with patients meeting individually with each sub-specialty provider; findings are then consolidated at a multidisciplinary case conference to generate an individualized rehabilitation plan. Care is guided and streamlined by a nurse practitioner with specialized training in brain injury, ensuring integrated assessment, clear communication across disciplines, and a unified set of recommendations for patients and families delivered the next day.

This review summarizes current evidence supporting multidisciplinary and targeted interventions including early introduction of sub-symptom threshold aerobic exercise, vestibular and physical therapy, vision therapy for oculomotor and accommodative dysfunction, cognitive rehabilitation, and behavioral health interventions; it also reviews symptom-targeted pharmacologic management when indicated. Using the MDCC as a clinical example, we also describe how this approach can be operationalized through coordinated examination-guided care. As the clinical evidence base continues to grow, there is broad consensus that care should be individualized, guided by the patient's symptom profile, and monitored over time, with the goal of delivering comprehensive care tailored to each patient.⁵ Recent guidelines highlight active rehabilitation, early education, and gradual return to cognitive and physical activity rather than strict rest, reflecting evidence that prolonged inactivity may exacerbate symptoms through deconditioning, autonomic dysregulation, and anxiety amplification.^{6,7} Physical examination findings can guide targeted rehabilitation and help tailor treatment to support recovery and return to daily activities, school, and sport.⁸

Key areas of multidisciplinary care rehabilitation

1. Education and Self-Management

At Boston Children's Hospital, education and collaborative management are the foundation of our MDCC model. We prioritize clear, practical guidance that helps families understand expected recovery trajectories, recognize warning signs that warrant urgent evaluation, and pace daily activities in a way that supports steady symptom resolution. We emphasize early, structured re-engagement with school and life through gradual, stepwise increases in cognitive and physical demands. Because recovery unfolds in the context of a child's home and learning environment, we actively engage caregivers and coordinate with schools to strengthen return-to-learn and return-to-work planning, an approach associated with reduced anxiety and improved functional outcomes.⁶

2. Neurological Assessment

Neurological assessment in the multidisciplinary concussion clinic focuses on identifying neurologic contributors to PPCS, clarifying headache phenotype, assessing risk factors for prolonged recovery, and determining whether additional diagnostic evaluation is warranted. The clinical history includes detailed review of the mechanism of injury, initial symptom burden, recovery trajectory, prior concussions, preexisting neurologic and psychiatric conditions, sleep disturbance, migraine

history, learning differences, chronic pain, and other factors that may influence recovery. Particular attention is given to risk factors associated with prolonged symptoms, including prior concussion, personal or family history of migraine, preexisting anxiety or depression, sleep dysregulation, chronic headache disorders, and autonomic vulnerability.

A central part of the neurologic evaluation is characterization of headache. The assessment aims to identify specific phenotypes and contributors including post traumatic migraine, tension type headache, occipital neuralgia, medication overuse, and headache exacerbated by sleep disorder or cognitive exertion. The neurologic assessment also considers cognitive symptoms such as inattention, slow processing, memory difficulties and executive dysfunction, particularly when they interfere with school performance and daily functioning.

The neurologic examination evaluates for focal deficits, gait and balance abnormalities, cervical findings and features that suggest an alternative diagnosis. These findings are integrated with evaluations from other specialists to guide symptom targeted management strategy. When cognitive symptoms are prominent, the neurologist integrates the clinical history with neuropsychological findings, school concerns, sleep and mood assessment and the boarder multidisciplinary evaluation.

Management is individualized and may include school accommodations, treatment of headache or sleep dysfunction and in selected cases pharmacologic treatment. Evidence for stimulant use in persistent neurocognitive symptoms after concussion remains limited and no standard pharmacologic approach is established. In selected cases, stimulants may be considered with careful consideration of risks, benefits, and alternative contributors to cognitive symptoms.⁹

3. Sub-Symptom Threshold Aerobic Exercise

As part of the MDCC, the Sports Medicine clinician assesses activity tolerance and typically prescribes graded, non-contact exercise. Equally important is identifying contributors to ongoing symptoms that may be distinct from the brain injury itself. For example, persistent headache is often labeled "post-concussion," but careful evaluation may suggest post-traumatic occipital neuralgia or a cervicogenic headache. Recognizing these patterns directs management toward targeted treatment, including physical therapy, acupuncture, massage or other modalities, and in selected cases occipital nerve blocks or trigger point injections, which can meaningfully reduce persistent symptom burden.

The Consensus Statement on Concussion in Sport recommends avoiding strict rest or isolation and introducing exercise early in recovery.³ Within 24 to 48 h, patients may begin light to moderate, non-contact activity with low risk of accidental contact, often starting with linear exercise such as stretching and walking. As symptoms stabilize, intensity and duration can be increased, while maintaining non-contact activity until full recovery is confirmed.

Brief, mild symptom increases are acceptable during exercise, defined as no more than a 2-point rise on a 0 to 10 scale with return to the pre-exercise symptom level within one hour after stopping. A practical benchmark we share with families is functional: if the patient can complete a similar amount of schoolwork one hour after an exercise session as they could have completed beforehand, that session is considered successful.^{10,11}

As tolerance improves, particularly with directional changes, we progress patients beyond linear activities to more complex, sport-relevant movement patterns while maintaining non-contact status. Re-engaging familiar motor skills such as non-contact skating for hockey, shooting for basketball, or ball-handling and cone drills for soccer can be therapeutic both psychologically and socially, and can also serve as graded vestibular rehabilitation.¹²

Aerobic exercise prescription and monitoring is a core component of multidisciplinary concussion rehabilitation and a key focus of coordinated clinic-based care.¹³ Early aerobic exercise is favored over strict rest, with multiple high-quality studies showing that active

rehabilitation shortens recovery time and reduces persistent symptoms compared with prolonged rest.¹⁴ Sub-symptom threshold aerobic exercise is structured activity performed below the level that provokes symptom exacerbation and has among the strongest evidence of non-pharmacologic interventions, with small to moderate improvements in symptom severity.¹⁵ Proposed mechanisms include improved autonomic regulation and cerebral blood flow.⁸ Across studies, outcomes are best when exercise is individualized and progressed thoughtfully, supporting both faster recovery and improved function.¹⁶ In our clinic, we encourage patients to return to safe physical activity as early as feasible and provide clear education on how graded exercise can reduce symptoms, improve sleep, prevent deconditioning, and lower the risk of autonomic dysfunction.

4. Vestibular and Oculomotor Rehabilitation

In our program, an Otolaryngology provider delivers a comprehensive evaluation and management plan focused on dizziness, balance impairment, tinnitus, hearing changes, and temporomandibular dysfunction. Persistent dizziness after concussion is multifactorial. Lightheadedness with standing or exertion, often described as post-traumatic exertional dizziness or hemodynamic intolerance, frequently responds to sub-symptom threshold aerobic training and reconditioning, along with optimization of hydration and electrolyte intake. Brief, episodic dizziness or vertigo, particularly when triggered by rapid head movements, may reflect vestibular impairment or centrally mediated vestibulopathy, both common after concussion and often responsive to vestibular physiotherapy, sometimes in conjunction with oculomotor and/or vision therapy when indicated. Screening can be performed with vestibular ocular motor screening (VOMS), but when vestibular deficits remain unclear, formal vestibular testing with a specialist may be warranted, including videonystagmography, rotary chair testing, vestibular evoked myogenic potential testing, calorics, video head impulse testing, and computerized dynamic posturography.¹⁷

When vertigo is brief and consistently triggered by head turning in a particular direction, especially when lying down, benign paroxysmal positional vertigo (BPPV) should be considered. BPPV can be diagnosed with office-based maneuvers such as the Dix-Hallpike and treated with canalith repositioning maneuvers such as the Epley maneuver, typically through a vestibular physiotherapist or other vestibular specialist. We have found BPPV to be a common and often under-recognized contributor to persistent post-concussive vertigo, sometimes going undiagnosed for months.¹⁸ When dizziness episodes are more prolonged, particularly when accompanied by headache exacerbations or other migrainous symptoms, post-traumatic vestibular migraine should be considered; it is a leading cause of persistent episodic vertigo in children and adolescents and often responds to lifestyle optimization and migraine-directed therapy.¹⁹

More pervasive daily dizziness persisting for months, particularly when accompanied by chronic imbalance, fear of falling, activity avoidance, and symptom provocation by postural changes, head movements, and visual flow stimuli (e.g., walking through a grocery store), may suggest persistent postural perceptual dizziness (PPPD or “3PD”). This functional disorder often arises in the setting of another dizziness driver (e.g., concussion or vestibular migraine), frequently co-occurs with anxiety, and typically responds to habituation-based vestibular physiotherapy, cognitive behavioral therapy and/or biofeedback, and serotonin reuptake inhibitor and/or serotonin norepinephrine reuptake inhibitor medications.²⁰

When vestibular symptoms after head injury are accompanied by hearing loss or persistent tinnitus, particularly if unilateral, further evaluation by an Otolaryngologist specializing in vestibular disorders should be considered, including audiometry and vestibular testing. In rare cases, these symptoms may reflect specific inner ear conditions requiring specialized management, including perilymphatic fistula, superior semicircular canal dehiscence, labyrinthine concussion, or

enlarged vestibular aqueducts.²¹ For patients with dizziness, balance disturbance, or visual motion sensitivity that persist beyond 10–14 days, vestibular therapy is recommended and can reduce vestibular symptom burden.²²

Temporomandibular dysfunction (TMD) may be triggered by chronic sympathetic nervous system activation in patients with PPCS, as well as with other causes of chronic dizziness and/or pain. We have noted it to be a frequent contributor to symptoms of tinnitus, subjective hearing impairment, ear fullness, and temporal headaches in our MDCC patients, and it can often be managed effectively with a 2-week non-chewing diet concurrently with a similar duration course of round-the-clock naproxen or ibuprofen.²³ Treatment-resistant cases may benefit from a night guard, physical therapy, and/or consultation with an oral surgeon specialized in temporomandibular joint disorders and/or orofacial pain.²⁴

5. Rehabilitation for Cervical and Other Musculoskeletal Symptoms

In our MDCC, each patient completes a comprehensive neurologic evaluation by a neurologist or sports medicine physician, along with a structured musculoskeletal assessment by a physical therapist with expertise in concussion management. The neurologic exam includes targeted screening for point tenderness in the neck, scalp, and trapezius, assessment of range of motion, and features that suggest a musculoskeletal contribution to symptoms.

Musculoskeletal symptoms, particularly neck pain, occur in 7 to 69% of patients in the acute post-concussion phase and increase to 90% in those with PPCS.²⁵

Our physical therapists screen for musculoskeletal contributors to pain and treat what they can reproduce and measure. The physical therapy assessment includes range of motion with symptom provocation, joint mobility when motion is limited, screening for whiplash associated disorders, sensorimotor testing such as joint position error, tests that stress cervical afferents such as smooth pursuit neck torsion, deep neck flexor endurance and motor control testing, and palpation of key muscle groups such as the suboccipitals and trapezius muscles when these reproduce symptoms.²⁶

Cervical musculoskeletal dysfunction commonly co-occurs with vestibular and vestibulo-ocular impairment after concussion, and this combined burden can drive a broad symptom constellation that includes headache, neck pain, dizziness, imbalance and instability, fatigue, exercise intolerance, and prolonged recovery. Physical therapy addresses vestibular complaints with an impairment-based, exercise-driven approach tailored to the exam, combining oculomotor and gaze stabilization training, balance and gait retraining to restore sensory integration, and habituation strategies for motion sensitivity. When benign paroxysmal positional vertigo is suspected, therapists confirm the diagnosis with positional testing and treat directly with canalith repositioning maneuvers such as the Epley or roll maneuvers. Gaze stabilization is progressed in a structured, symptom-limited fashion to improve visual clarity with head or target movement and support return to daily function.²⁷

Another important area for physical therapy is autonomic symptoms and exercise intolerance.²⁸ When exertion reliably worsens symptoms beyond the acute stage, physical therapists evaluate for physiologic dysregulation reflected in abnormal heart rate and blood pressure responses to position change and low-intensity activity, which can trigger dizziness, lightheadedness, and headache. Graded exertional testing, including baseline hemodynamics and the Buffalo Concussion Treadmill Test, is commonly used to define a symptom-limited threshold and guide a structured aerobic prescription. Patients are then started on sub-symptom threshold aerobic exercise at roughly eighty percent of the peak heart rate achieved on testing, with close heart-rate monitoring, brief daily sessions most days of the week, and gradual progression with warm-up and cool-down to reduce symptom flares.

Physical therapists are central to concussion care, providing

education, guiding safe return to activity, and treating impairments that limit return to school, sport, and daily function. PT helps by translating examination findings into a structured, progressive rehabilitation plan that targets the specific impairments driving symptoms and participation limits. An individualized, goal-directed physical therapy approach is an essential part of the multidisciplinary approach to optimize outcomes and support a timely return to full participation, and to limit the need for pharmacological intervention and costly tests.²⁹

6. Cognitive Sequelae and Neuropsychological Management

Cognitive symptoms are common and clinically impactful among the 25–35% of children who develop PPCS, defined as symptoms extending beyond 28 days post-injury.^{30–32} Although severe or global impairment is not typical, problems with attention, executive function, and memory are observed acutely, with some literature documenting relative weaknesses on objective, performance-based measures of function at 1 month post-injury.^{33,34}

Individual recovery trajectories are highly variable, but data suggest that cognitive symptoms rarely occur in isolation and often cluster with emotional symptoms and fatigue.³⁵ Further, risk for prolonged symptoms is higher with female sex, adolescence, preinjury mental health concerns, and adverse family/contextual factors.³⁶ Girls have nearly twice the odds of chronic PPCS. While evidence suggests adolescents improve more slowly than younger children, a recent investigation indicated 62% of very young children (<72 months) exhibit cognitive symptoms at 1-month post-injury according to caregivers.^{37,38} In adolescents and school-aged children, PPCS substantially affects health-related quality of life across physical, emotional, social, and school domains, with school functioning representing one of the most enduring challenges.^{39,40}

In our MDCC, the role of neuropsychology is to analyze and interpret injury characteristics, emotional functioning, and contextual factors contributing to a patient's presentation. These data are integrated with objective test data to clarify diagnosis and guide targeted intervention. Brief, focused assessment examines the cognitive domains most vulnerable to head injury, including attention, executive functioning (e.g., working memory, processing speed), and learning and memory.^{33,34,41} We obtain information about social, emotional, and behavioral functioning with standardized rating scales.

Findings are integrated with developmental, medical, psychosocial, and academic histories to help determine the predisposing, precipitating, and perpetuating factors that contribute to PPCS.^{36,42–44} Predisposing factors include psychiatric morbidity (e.g., anxiety or depression), prior concussions, neurodevelopmental disorders (e.g., ADHD, learning disorder), personality style, and broader social context. Precipitating factors refer to the biomechanical injury itself, acute symptom burden, and immediate stress, pain, and/or patterns of rest or activity that follow.⁴⁵ Perpetuating factors encompass the processes that maintain or amplify symptoms over time (new/worsening anxiety/depression, catastrophic interpretations, avoidance/prolonged rest, sleep disruption, chronic pain, environmental/academic stressors). Biopsychosocial and cognitive behavioral therapy (CBT)-informed models consistently emphasize these perpetuating influences as key drivers of prolonged recovery.⁴⁶

In collaboration with other sub-specialty providers, neuropsychology works with the lead nurse practitioner to help the team prioritize and tailor non-pharmacologic treatments endorsed in PPCS pathways, such as developmentally appropriate education, reassurance, and CBT-informed interventions for mood, anxiety, and maladaptive avoidance. Further, the neuropsychologist provides specific return-to-learn guidance, including temporary workload reduction, rest breaks, environmental modifications, and extended time.^{1,47,48} Last, neuropsychological findings also help guide medication decisions when indicated, particularly when targeting executive dysfunction and/or mood symptoms that are contributing to prolonged recovery.

By integrating targeted assessment, interdisciplinary consultation, and brief intervention within a shared clinic visit, neuropsychology becomes essential to the efficient, high-quality multidisciplinary management of PPCS.

7. Sleep-Focused Recovery Planning

Sleep counseling is integrated into every patient visit in our MDCC, with a focus on identifying modifiable drivers of poor sleep such as irregular schedules, prolonged daytime naps, evening screen exposure, caffeine timing, pain, and stress. Sleep is assessed using the Patient-Reported Outcomes Measurement Information System (PROMIS) sleep score, a standardized patient-reported measure of sleep quality and sleep-related functioning derived from validated item banks. PROMIS includes Sleep Disturbance and Sleep-Related Impairment short forms that support efficient assessment and longitudinal tracking. By the end of the visit, the neurologist provides a consolidated set of recommendations that pairs practical sleep hygiene guidance with individualized adjustments to routines and symptom management, coordinated with return-to-learn supports and school accommodations when needed, and referrals for evidence-based outpatient therapy when sleep disruption is maintained by comorbid anxiety, mood symptoms, or maladaptive sleep behaviors.

Sleep counseling is classified as “likely to be beneficial” when sleep disturbance is present after mild traumatic brain injury.⁴⁹ The American Medical Society for Sports Medicine recommends addressing sleep hygiene and monitoring sleep in athletes with post-concussion sleep disturbance, using nonpharmacologic and, when indicated, pharmacologic strategies.²⁷ In a pediatric study, the largest symptom improvement occurred following the session focused on concussion psychoeducation and sleep hygiene.⁵⁰

CBT is often recommended because evidence supports benefit for persistent sleep difficulty after concussion. In a randomized controlled wait-list trial of telehealth-delivered CBT for insomnia in adults at least four weeks post-concussion with clinically significant insomnia, CBT produced greater improvements in insomnia severity and sleep quality than wait-list control, with benefits maintained for 6 to 12 weeks, and greater sleep improvement was associated with reduced PPCS burden.⁵¹

8. Ophthalmology and Optometry Assessment

MDCC patients undergo a comprehensive evaluation by both an optometrist specialized in oculomotor assessments and an ophthalmologist. The assessment includes screening of visual acuity, pupillary function, visual fields, ocular alignment and motility, vergence, accommodation, visual tracking, refraction, eye pressure, and examination of ocular adnexa, orbits, and anterior and posterior segments. This thorough approach helps identify treatable contributors to vision-related symptoms such as blurred vision, diplopia, eye strain, light sensitivity, headaches, difficulty with reading, and visual fatigue. An initial study of 57 MDCC patients showed that 72% had difficulty with their eyes when reading, 50% blurred vision, 26% light sensitivity, and 21% eye fatigue/strain.⁴ Another study of the 140 patients seen in the MDCC between 2017 – 2019 showed that 42% had deficits in vergence, 70% had deficits in accommodation, and 43% had visual tracking deficit.⁵² A recent prospective study further confirmed that nearly 80% of concussed children experience measurable vision deficits in vergence and accommodation that persist for months without targeted intervention.⁵³

Screening PPCS patients for vision-related symptoms should be done objectively. First, the Convergence Insufficiency Symptom Survey (CISS)⁵⁴ is a 15-item questionnaire completed by the MDCC patients at intake, identifying objectively symptoms like reading difficulty, eye strain, headaches, blurry vision, and double vision. It includes performance, somatic, and vision-related items, which makes it useful for understanding the patient's day-to-day functional impact. The survey,

though not validated for concussion patients, systematically documents symptoms in this cohort.⁵⁵ A retrospective study of 210 pediatric and adolescent patients, that included patients seen in the MDCC, revealed higher vision-related symptoms in those with vergence or accommodation deficits, supporting its use as a screening tool.⁵⁵ Further, our recent prospective study in children 8 to 17 years old showed that in adolescent concussed patients who had vergence and/or accommodation deficits, the symptom reporting was higher on the CISS for vision and performance related sub scores compared to those without such deficits. Elevated symptom reporting was evidence of increased subjective discomfort when doing near work.⁵⁶ Second, a recent infographic on screening tools which are near point of convergence and amplitude of accommodation for common concussion-related vision disorder suggests the exact methodology with a video reference on how to test vergence and accommodation. The findings from these two tests along with the CISS scores can be used to refer patients to a specialist who can do detailed assessment of the oculomotor system.^{55,57}

Based on the symptom reporting and findings during the detailed optometric examination, the management course is first aimed at accommodations that can help with better visual functioning. In practice, this starts with reducing visual load and environmental triggers such as bright light and prolonged screen use, shortening near work, and using targeted academic supports when reading or sustained visual attention worsens symptoms, including simplified visual materials, access to preprinted notes, recorded lectures, and audiobooks or text-to-speech programs. When blurred vision or diplopia is driven by accommodative or vergence dysfunction, appropriate refractive correction for distance or just near can lessen symptoms and improve function while recovery is underway. Even small amounts of refractive error, not normally prescribed, may help a patient during recovery phase.⁴ Horizontal or vertical prisms are considered on a case-by-case basis depending upon the etiology of diplopia. Light sensitivity, when debilitating or persistent, is managed with low-tinted lenses but with caution to avoid patient dependency due to neural adaptation; these lenses are prescribed only when symptoms significantly impact daily functioning. While earlier consensus statements questioned isolated vision therapy effectiveness post-concussion⁵⁸, recent randomized trial data in adolescents and young adults with concussion-related convergence insufficiency support office-based vergence and accommodative therapy with home reinforcement as an effective intervention that can shorten the symptomatic period when initiated promptly.⁵⁸ In MDCC patients with vergence and accommodation deficits, home-based exercises are prescribed when necessary. Even when clinical findings are normal, high symptom reporting or symptom provocation is an avenue to consider for vision rehabilitation.⁵⁶ Habituating the visual system with vision-rehabilitation to reduce symptom provocation helps with recovery in these persistently symptomatic concussed patients. Finally, vestibulo-ocular dysfunction often co-occurs with many oculomotor dysfunctions⁵⁹, we view vestibular physical therapy and vision-focused rehabilitation as complementary and use an impairment-based approach to match referrals and progression to each patient's symptom profile and functional goals. Our MDCC post-clinic conference among the multidisciplinary providers identifies overlapping therapies and adjusts the prescribed therapies with the goal of individualizing care and avoiding repetitive rehabilitation that takes time from other activities.

Along with the comprehensive oculomotor assessment provided by the optometrist, a thorough ophthalmic assessment is performed by our ophthalmologist. First, since diplopia is common vision symptom in PPCS, cranial nerve III, IV, and VI palsies must be ruled out by a detailed sensorimotor evaluation including the measurement of eye alignment in multiple directions of gaze, ocular motility assessment, and sensory assessment for stereopsis, fusion, and diplopia. Although rare,^{58,60} the presence of cranial nerve palsy necessitates neurological imaging⁶¹ with computed tomography to rule-out subtle fractures or magnetic resonance imaging to track the course of the relevant cranial nerves observing for disruption or hemorrhage may be important to guide

treatment towards the underlying etiology when appropriate. Treatment for these conditions may include observation and prism therapy initially. Definitive therapy with eye muscle surgery may be considered in unmitigated cases.⁶⁰

Second, an assessment of the anterior and posterior segments of the eye must be performed. In the anterior segment, evaluation of the ocular surface is critical as a smooth tear film maximizes vision and minimizes visual discomfort and photophobia. Although not studied in the pediatric population, a retrospective study of US veterans showed a higher prevalence of dry-eye syndrome in those with all traumatic brain injury.⁶² The same study showed that the association with ocular pain was especially prominent after traumatic brain injury.⁶² Further, a prospective study of veterans with a small cohort of non-blast traumatic brain injury showed low tear production in 22%, elevated tear osmolality in 67%, and ocular surface staining in 78%.⁶³ The ocular surface staining was significantly different than control suggesting that traumatic brain injury patients had more disease. Since dry eye disease may lead to blurry vision, monocular diplopia, asthenopia, and fatigue with reading, many symptoms that PPCS patients describe, our evaluation looks for signs of ocular surface dryness and attempts to incorporate its treatment into management. We will recommend supplemental artificial tears, and we will also recommend eyelid hygiene (warm compresses applied to the eyelids once daily for 30 seconds followed by a gentle soap scrub of the eyelids and eyelashes once daily) for any patient with eyelid anomalies that may be affecting the tear film. In addition to the ocular surface, an assessment of the cornea, anterior chamber, iris, and lens is critical to ensure no anomalies appear in those structures, either pre-existing or associated with indirect injury.

In the posterior segment, evaluation of the vitreous, optic nerve, and retina is paramount. The vitreous and its adhesions to the retina anteriorly at its base as well as to the optic nerve, macula and retinal vasculature may lead to hemorrhage or retinal tears in the pediatric population. These findings on examination may associate with symptoms such as floaters, shadows in vision, flashing lights, or parts of the visual field being blurry. Examination may identify debris or hemorrhage within vitreous, tears in the retina, and even detachment or dialyses of the retina. Rarely, the optic nerve may be damaged especially in the intracanalicular portion through indirect shearing mechanisms from head injuries.^{61,63} This finding may be associated with symptoms of diminished visual acuity and visual field. Examination may show optic nerve head show pallor and depression of the visual field locally or globally. In addition, thinning of the retinal nerve fiber layer may also be seen at least based on adults with concussion.^{64,65} Thus, our evaluations look for signs of vitreous hemorrhage, retinal injury, and optic nerve compromise; when found, we further classify them with retinal and optic nerve imaging and occasionally perform visual field testing to document the injury and to track its progression over time.

9. Symptom-Targeted Pharmacological Management in MDCC

At MDCC, medication decisions are made at the end of the visit in consultation with the multidisciplinary team, with treatment focused on the symptom that is most limiting function for each patient, and management of headaches, anxiety, and cognitive dysfunction is often directed by the neurology team. Pharmacologic treatment in concussion care is symptom targeted and adjunctive, not primary therapy.⁶⁶ No medication is FDA approved specifically for concussion, and current consensus supports using medications selectively for persistent, function-limiting symptoms while prioritizing active rehabilitation and a careful return to activity. Medication selection is matched to the dominant symptom pattern, such as phenotype-guided management of post-traumatic headache, use of melatonin or magnesium glycinate to support sleep, headache preventive therapy when clinically indicated, and selective consideration of agents for fatigue, attention, or mood symptoms in patients with persistent symptoms. No pharmacologic approach has sufficient evidence-based support as a first-line,

stand-alone treatment; medications are best used within a multidisciplinary plan that includes aerobic exercise, targeted rehabilitation, behavioral strategies, education, and close follow-up. When medications are used, they are initiated at low doses and titrated slowly, with careful monitoring for adverse effects and medication overuse.

Across 23 studies evaluating 20 interventions in 8,277 patients, 78% of studies reported some symptom improvement, yet the authors concluded that no intervention had adequate data to justify routine frontline use in practice.⁶⁷

Table 1 summarizes the medications most commonly used in our clinic, organized by symptom domain and clinical indication, to illustrate how symptom-targeted pharmacotherapy is integrated as an adjunct to active rehabilitation within the MDCC care model.

10. The Boston Children's Hospital Multidisciplinary Concussion Clinic

The MDCC is a single coordinated clinic, where the services summarized in Table 2 are co-located. It provides a single-day evaluation for patients experiencing PPCS spanning multiple symptom domains, allowing for appointments with all providers in the same day to prevent fragmented care and allowing a collaborative approach to treatment. Patients are accepted from both Boston Children's Hospital providers and external clinicians.

Referrals to the MDCC come from both internal and community clinicians, most commonly from the Boston Children's Hospital Neurology and Sports Medicine departments, with a smaller number from outside primary care and other referring providers. Patients are typically selected for MDCC evaluation when they have moderate to severe persistent post-concussion symptoms involving multiple domains and have not improved with initial first-line management in the community or within standard specialty follow-up.

Upon receipt of a referral, the MDCC nurse practitioner, who also practices in the Department of Neurology at Boston Children's, reviews the clinical information and serves as the clinic's coordinator. She conducts a triage assessment to determine whether the patient is likely to benefit from the multidisciplinary model, including confirmation that

symptoms are attributed to concussion, assessment of whether the symptom presentation aligns with the specialties represented in the clinic, and review of prior evaluations to determine whether the patient has recently been seen by similar providers within Boston Children's or externally.

Patients deemed appropriate for the MDCC are contacted and provided with a comprehensive intake form. The intake collects detailed information regarding the concussion, prior concussion history, past medical history, family medical history, and academic functioning. After completion of the intake process, patients are scheduled for evaluation. Most patients are seen within four weeks of referral.

On the day of evaluation, patients complete an updated intake assessment to facilitate efficient and focused clinical appointments. This assessment includes validated patient-reported outcome measures, such as PCSS, PROMIS, CISS, and the Headache Impact Test (HIT). These instruments are administered on the morning of the visit to provide clinicians with the most current information regarding symptom burden and functional impact.

Throughout the visit, the nurse practitioner serves as the central point of contact for patients and families, facilitates transitions between specialty appointments, and addresses questions that arise during and after the visit. At the conclusion of the clinical evaluations, she convenes a same-day multidisciplinary case conference with all participating providers to review each patient's presentation and develop consensus recommendations that establish an individualized plan of care.

Following the case conference, the nurse practitioner documents and finalizes the care plan and shares it with patients and families via Boston Children's patient portal, with telephone follow-up as needed for questions or clarification. For more complex needs, a scheduled phone call or virtual visit may be arranged. MDCC functions as a one-time consultative service; when ongoing specialty care is indicated, patients are provided with contact information for relevant clinics to arrange follow-up. Based on the multidisciplinary care plan, the nurse practitioner may also provide school accommodation letters, prescribe medications, or place orders for additional diagnostic testing.

With respect to functional outcomes and patient satisfaction, we acknowledge these data would further support the value and need for

Table 1
Medications commonly used in the management of persistent post concussion symptoms.

Medication Class	Common Indications	Examples	Additional Considerations
Migraine Preventatives	-Vestibular migraine -Post-traumatic headache	-Tricyclic antidepressants -SNRI -Propranolol -Topiramate -Cyproheptadine -CGRP antagonists (monoclonal antibodies or gepants)	-SNRI can also be helpful for patients with concurrent anxiety and/or PPPD -Propranolol can also be helpful for patients with concurrent anxiety, autonomic dysfunction and orthostatic dizziness
Migraine Abortives	-Vestibular migraine -Post-traumatic headache	-Triptans -NSAID -CGRP antagonists (gepants)	
SSRI/SNRI	-Anxiety -Depression -PPPD	-Venlafaxine -Duloxetine -Sertraline -Fluoxetine -Escitalopram	SNRI can also be helpful for patients with concurrent PPPD
NSAIDs	-Post-traumatic headache -Vestibular migraine -TMD	-Naproxen -Ibuprofen	Limit to prevent medication overuse and rebound headaches
Stimulants	-Cognitive/Attention Issues -ADHD	-Methylphenidate -Dextroamphetamine -Amantadine	
Supplements	-Post-traumatic headache -Vestibular migraine -Residual concussion	-Magnesium -Riboflavin -Coenzyme Q10	

Abbreviations: SNRI – serotonin norepinephrine reuptake inhibitor, CGRP – calcitonin gene related peptide, PPPD – persistent postural perceptual dizziness, NSAID – non-steroidal anti-inflammatory drug, SSRI – serotonin reuptake inhibitor, TMD – temporomandibular dysfunction, ADHD, attention deficit hyperactivity disorder.

Table 2
Clinical disciplines in the multidisciplinary concussion clinic.

Specialty	Key Assessment Targets	Primary Intervention
Neurology/Sports Med	Autonomic, Cervical, Headache. Review of cognitive testing	Diagnosis and symptom phenotyping, autonomic assessment and exercise prescription, recommendations for school accommodations, medication management for headache, anxiety, and executive dysfunction when indicated, coordination of return to school and return to sport, sleep management, referrals (including acupuncture), and ordering imaging studies when clinically warranted. Follow up planning.
Physical Therapy	Vestibular/ Balance, Cervical mobility,	Musculoskeletal intervention, manual therapy, Gaze stabilization, BPPV maneuvers, Balance retraining, Conditioning and habituation
Ophthalmology	Ophthalmic Structural Integrity, Assessment of Sensorimotor Evaluation	Comprehensive ophthalmic examination for ocular, adnexal, and orbital pathology with management; Partnership with optometry in treatment of sensorimotor imbalance
Optometry	Accommodation, Vergence, Tracking	Detailed oculomotor assessment of vergence, accommodation, and visual tracking, with management that may include refractive correction for distance and/or near, vision therapy to rehabilitate oculomotor skills, prism lenses when appropriate, tinted lenses for light sensitivity, and vision-related accommodations.
Neuropsychology	Attention, Executive Function, and memory Mood Social context	Cognitive pacing, CBT-informed coping, School accommodations
Otolaryngology	Inner ear/ Hearing, Peripheral vestibular	Medical management of vestibular migraine, persistent postural perceptual dizziness (PPPD), and/or temporomandibular dysfunction; vestibular/audiologic testing (when indicated); canalith repositioning maneuvers (when indicated)
Nurse Practitioner as Care Coordinator Leader		Provides summary to patients/ caregivers following the same-day case conference, medical provider that and answers patients/ caregivers' questions, central contact for school/work accommodations and follow-up

the MDCC model. At present, our feedback sample is limited and not yet sufficient to support meaningful analysis. However, recent workflow changes have substantially improved survey completion, and we expect to have a more robust dataset in the near term to enable formal evaluation for quality improvement and future publication.

Conclusion

The Multidisciplinary Concussion Clinic model is a clinically useful and patient-centered approach for children and young adults with PPCS. By bringing key specialists together in a single coordinated evaluation, this model improves efficiency for patients and families, reduces duplication of assessments, and minimizes conflicting treatment recommendations across disciplines. It also allows the team to identify the

most important symptom drivers, align care around shared priorities, and deliver a unified set of comprehensive follow-up recommendations that are practical, individualized, and easier to implement. By consolidating diagnostic reasoning and treatment planning into one coordinated pathway, this model saves time, reduces care fragmentation, and streamlines follow-up across rehabilitation, school supports, and symptom-targeted management. This model gives patients a single, coordinated entry point, reducing stress and time away from school and activities while keeping the treatment plan consistent across specialists. In this way, multidisciplinary care improves the overall care experience while supporting high-quality rehabilitation for complex concussion recovery.

Statement for submission

I have nothing to declare.

CRediT authorship contribution statement

Karameh Hawash-Kuemmerle: Conceptualization, Supervision, Writing – original draft, Writing – review & editing. **Jacob R. Brodsky:** Writing – review & editing. **Heather Molind:** Writing – review & editing. **Kate Mysak:** Writing – review & editing. **Michael O'Brien:** Writing – review & editing. **Jessica Pidgeon:** Writing – review & editing. **Aparna Raghuram:** Writing – review & editing. **Ankoor S Shah:** Writing – review & editing. **Rebecca Stevens:** Writing – review & editing. **Alex M. Taylor:** Writing – review & editing.

Declaration of competing interest

The author declares that there are no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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